

No Free Null: Reference Distributions for Explaining Reinforcement Learning Agents

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Abstract. Deciding whether an explanation of a reinforcement learning agent means anything requires a reference distribution: what it would have looked like had the agent learned nothing. Reinforcement learning has no agreed way to build one, and the problem is harder than it appears. The construction available off the shelf, randomising network parameters to generate reference draws, has very little power: across sixteen checkpoints on four control tasks it fires once, barely ($z = +3.00$), on the single fully converged agent, where an environment-level null reaches $z = +191.9$. Its width closely tracks the standard deviation of random-initialisation return, so it measures the variance of initialisation and not the explanation. Run instead in its canonical form it answers a different question and clears the explanation we show to be empty. We then build the three natural replacements and each fails differently. A signal-free corpus varies the corpus as well as the information. Blinding the observation channel removes the agent’s capacity to respond, collapsing the reference onto a point mass, so that on two of our three cases it cannot decide at all. Permuting the outcomes, the direct analogue of the supervised label-permutation test, leaves the price forecasting a label it no longer matches and hands the null agents an arbitrage worth \$49 an episode. In each construction we built, removing the information also perturbed something else, and the perturbation decided the verdict. We motivate this with an agent whose explanation is stable across seeds, semantically legible, and empty, and which can be driven from 39.9% to 3.2% of attribution mass at no detected cost in return.

1 Introduction

Interpretability is proposed as a basis for oversight: if we can see which inputs drive a policy, we can decide whether to trust it. That needs an attribution to mean something, and attribution methods return an answer whether or not it does. Given any policy and state, Shapley values [6, 9] and integrated gradients [11] produce a ranked, signed vector. Nothing in the output separates an attribution reflecting structure the agent exploits from one reflecting nothing.

Telling them apart needs a *reference distribution*: what the explanation would have looked like had the agent learned nothing. Supervised learning has two standard ones, both from Adebayo et al. [1]: randomise the parameters, or train on permuted labels. Reinforcement learning has adopted the first and not the second. Huber et al. [5] apply the parameter test to Atari saliency and observe that the data-randomisation test has no obvious RL analogue. The obstacle is structural rather than clerical: supervised labels can be permuted while the input distribution is held exactly fixed, whereas an RL agent’s observations, actions and trajectories are all generated by the policy, so there is no label to permute that leaves the rest of the problem untouched.

We set out to supply one. The result is negative and the negative result is the contribution: parameter randomisation, repurposed to generate reference draws, has almost no power, and each of the three natural replacements fails in its own way. Kept in its original form as a similarity check the parameter test remains sound, but it answers a different question and clears the explanation

we show to be empty.

Contributions.

1. Parameter randomisation is a near-powerless reference in RL, and a variance decomposition explains it: the span inherits the spread of random-initialisation return, which it tracks to within two percent on the tasks where either is non-degenerate.
2. Three replacements, three distinct failures, each with a measured mechanism.
3. A worked case of an explanation that passes every conventional check we could apply and carries no information.
4. Attribution is not identified by task performance: it can be steered at a cost that, in our controls, tracks whether the feature is load-bearing.

What we do not claim. The argument that RL attribution should target outcomes rather than policy outputs is due to Beechey et al. [3]; FastSVERL [2] addresses the scalability, off-policy and temporal problems we only name. Our test adapts Adebayo et al. [1]. Deletion-based fidelity is established practice in XRL [4, 7]. Our contribution is the RL construction, the demonstration that four references fail and why, and the consequences.

2 Method

Let $v(S)$ be the expected episode return of an agent observing only the features in $S \subseteq N$, with features

outside S replaced at every timestep by draws from a reference distribution. Define the *attribution span* $\Delta = v(N) - v(\emptyset)$.

Shapley values and integrated gradients satisfy additivity identities under which per-feature attributions sum to Δ , so the span can be read off either. It is worth stating that Δ is *not* a Shapley quantity: it is the difference between two masked rollouts and depends only on the masking. This has a practical consequence: it costs two evaluations rather than $2^{|N|}$.

The statistic needs a reference distribution but does not say how to build one, which is this paper’s subject. Every construction we try corrupts the observation channel somewhere between corpus and agent (Figure 1), differing in where and so in what else it disturbs. Their failures give three requirements: preserve the observation stream, the agent’s responsiveness, and the difficulty of the task. Each construction below sacrifices exactly one.

Given null draws with mean $\bar{\Delta}$ and standard deviation s we report a rank statistic and $z = (\Delta_{\text{obs}} - \bar{\Delta})/s$, and require both to reject. We write the first as a rank rather than a calibrated p -value: an exact randomisation p -value needs exchangeability under the null, which is what this paper puts in doubt. The rank statistic assumes no shape but bottoms out at $1/(n+1)$; the normal statistic has resolution but assumes a shape the null need not have. If every null draw is identical, $s = 0$; we return $z = \pm\infty$ and surface the degeneracy rather than guarding it away, but treat it as a reason to distrust the reference. A point-mass null has no specificity, and building one is how a more principled-looking null construction turned out to fire on a corpus with no signal in it (Section 4.6).

3 Setup

Environments with known answers. Explanation evaluation lacks ground truth [4]. We construct fixed-length synthetic corpora where the answer is fixed by construction: a *planted signal* (one of 18 features informative about the outcome), a *null* (none), and a *decoy* (one feature correlating 0.989 with the outcome in-sample and 0.006 out of sample).

Market. Our main setting is a binary contract settling to \$1.00 if BTC is higher after 15 minutes, on a regulated prediction market: 6,425 settled contracts over 68 days, walk-forward split by time, test evaluated once. Every episode resolves to a known value. Agents are trained with PPO [8]; reward is the change in mark-to-market equity net of the exchange’s published fee schedule, a 9% round-trip hurdle at the price this contract trades at.

Control tasks. CartPole-v1, Acrobot-v1, MountainCar-v0 and Pendulum-v1, whose observation spaces are small enough to enumerate all coalitions, so Shapley values there are exact. Pendulum’s continuous action is discretised into nine values so the same

Table 1: Test split, 1,284 held-out episodes. Paired bootstrap against always-flat.

Policy	Mean P&L	Trades/ep	p vs flat
always-flat	+0.000	0.00	n/a
buy-and-hold	−1.608	1.00	0.23
PPO (5 seeds)	−0.595 ± 0.176	1.37	0.10
mean-reversion	−11.615	4.94	<0.0001
random	−18.532	9.33	<0.0001

Table 2: Null-model test. Null from 24 blind-trained agents, span $+8.19 \pm 5.11$.

Case	Span	z	Verdict
planted signal	+70.89	+12.27	informative
null corpus	+11.86	+0.72	not distinguishable
real market	+9.37	+0.23	not distinguishable

categorical PPO trains on every task.

4 Results

4.1 An explanation that passes every check

The market is empirically efficient: its implied probability has a weighted mean absolute calibration error of 0.0172, and spot return correlates 0.476 with the outcome but only 0.016 with the residual after the price is accounted for. No policy beats abstention (Table 1).

PPO’s shortfall is its own: on synthetic data with no signal by construction the same agent scores -0.47 ± 0.47 . Yet across five seeds whose pairwise performance differences are indistinguishable in all ten pairs, its attributions have cross-seed rank correlation 0.850 with 100% agreement on the top feature, a top-5 ranking whose adjacent gaps exceed their combined standard errors, and a legible story: time-to-expiry dominates, reading as an agent that learned when not to trade.

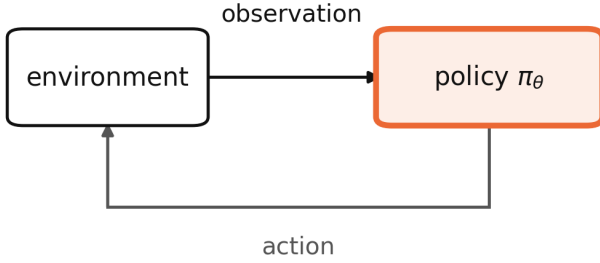
The null test disagrees (Table 2, Figure 2). It has power and specificity, and the market agent sits at the centre of the distribution of explanations of nothing.

We note one consequence for our own reporting. An earlier draft stated that “the whole observation is worth +5.914 per episode against being blind” and presented it as a finding. Blind agents produce $+8.19 \pm 5.11$. It was noise.

4.2 A positive control on real data

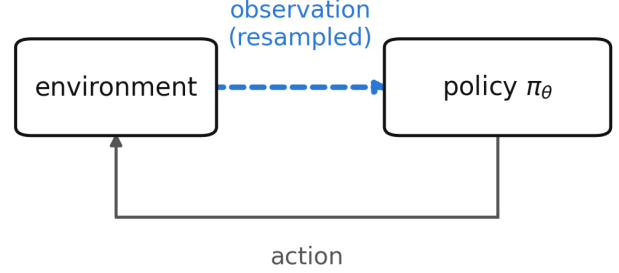
Every case above where the test fires is synthetic, which invites the objection that it detects planted signal rather than information. Holding the corpus, features, normalizer, split and null construction fixed and varying only the objective answers it. The market’s price is calibrated to 0.0172, so it is highly informative *about the outcome* while offering nothing to trade against once a 9% round-trip fee is paid. Scoring the same architecture on the same real episodes for *calling* the settlement rather than trading it, against a null of 24 agents trained on those episodes with observations blinded, gives a span

(a) parameter randomisation



weights resampled, so the policy never learned anything.
masking its inputs moves return arbitrarily.

(b) observation corruption (ours)



the policy is trained normally on a real reward.
the only thing withheld is information.

Figure 1: Two ways to build a reference distribution of explanations for an agent that learned nothing. The established check (a) randomises the policy’s parameters, leaving a network that was never trained. Ours (b) corrupts the environment’s observation channel instead, so the agent optimises a real reward with nothing to condition on. Only (b) yields a normally trained policy, which is the failure mode that occurs in deployment.

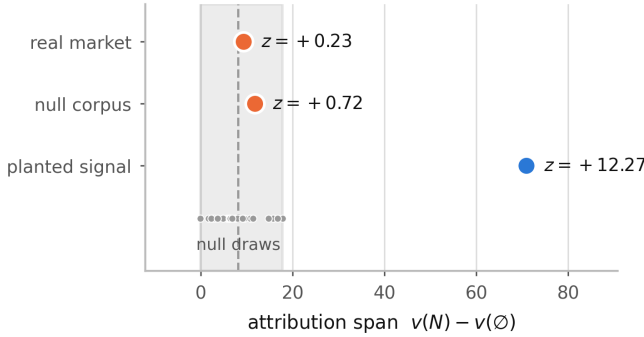


Figure 2: Attribution spans against the null. The band is the range over 24 agents trained where the observation channel carries no information.

of $+7.46$ against -0.48 ± 2.34 : $z = +3.40$, informative, unchanged in all 10,000 resamples. The trading agent on identical episodes and an identical null gives $+7.41$ against $+2.47 \pm 2.29$, $z = +2.16$, which the two-part rule declines, though only 66% of resamples agree and we report it as borderline. The two spans are nearly the same size; what separates them is the null, since a blind agent cannot call the outcome at all but can still decline to trade. The market data is not uninformative. The information is not monetisable, and an attribution of the trading agent cannot tell those two situations apart.

4.3 Validation against ground truth

On the planted corpus, outcome attribution ranks the planted feature first with 55% of total mass. On the decoy corpus, an agent trained where the decoy predicts the outcome (in-sample $+47.24$, held out -6.83) is explained on episodes where the decoy is provably worthless: per-decision attribution gives it 6.5% of mass at rank 3 of 18, outcome-based attribution 1.1% at rank 10. Both are true; the policy does key on the decoy. Per-decision attribution is correct about behaviour and misleading about value.

4.4 Parameter randomisation is an underpowered reference

Randomising parameters is used two ways. Adebayo et al. [1] degrade a single explanation and ask whether it changes. Here we use it the other way, as a source of reference draws against which a trained agent’s span is placed, which is the construction available to a practitioner wanting an RL null. The two answer different questions, and in the full paper we run the canonical version as well: it clears the explanation we show to be empty, on both explanatory targets we tried, so passing it is not evidence that there is anything to explain.

We build both nulls for the same task (Table 3, Figure 3). A randomly initialised policy behaves arbitrarily, so masking its inputs moves return unpredictably and the reference distribution is very wide. Across sixteen checkpoints the parameter null fires once, barely ($z = +3.00$), on the converged CartPole policy scoring 500 of 500, where the environment null reaches $z = +191.9$. The observed span is exactly zero at twelve of the sixteen, so only CartPole tests anything. The two return the same verdict on nine: four on Pendulum where both decline on a zero span, four on Mountain-Car where neither can decide, and the converged CartPole agent where both fire. Of the four non-zero spans they disagree on three. The parameter null is therefore not powerless, but its power appears at convergence and nowhere below it, which five independently trained agents per task confirm.

Why it is wide. The gap has a mechanism, and it is not about explanations. On every task the parameter null’s width closely tracks the spread of random-initialisation return: 140.43 against 140.86 on CartPole and 110.59 against 109.32 on Acrobot, both inside two percent, and 127.40 against 138.88 on Pendulum, inside nine. In $\Delta = v(N) - v(\emptyset)$ the masked term is near-constant for a random network (sd 1.8% of the unmasked

Table 3: Null construction on four control tasks, 24 draws each. MountainCar is degenerate under both: no policy in the budget reaches the goal. Pendulum’s agent has a span of exactly zero at every checkpoint, so it constrains the widths but supplies no detection evidence.

Environment	Parameter	Environment	Ratio
CartPole	$\pm\mathbf{140.86}$	$\pm\mathbf{2.52}$	$56\times$
Acrobot	$\pm\mathbf{109.32}$	$\pm\mathbf{0.00}$	degenerate
MountainCar	$\pm\mathbf{0.00}$	$\pm\mathbf{0.00}$	$0\times$
Pendulum	$\pm\mathbf{138.88}$	$\pm\mathbf{27.50}$	$5\times$

Table 4: Steering one feature’s attribution. Every figure in a row comes from one run at invariance penalty $\lambda = 20$, fixed in advance. p is a cluster bootstrap against the $\lambda = 0$ agent, resampling seeds and then episodes within seeds.

Corpus	Attribution	Return	p
market	39.9% $\rightarrow$ 3.2%	$-3.49 \rightarrow -1.65$	0.32
planted signal	46.5% $\rightarrow$ 1.5%	$+45.04 \rightarrow +\mathbf{8.66}$	<0.001

term’s on CartPole), because a policy not using its observations loses little when they go. The span inherits the unmasked variance entire: *parameter randomisation measures the variance of initialisation*, which is why a test built on it has almost no power here. Blinding draws each coordinate independently, which on Pendulum satisfies $\cos^2 + \sin^2 = 1$ for only 0.5% of draws. Resampling whole observation rows instead, which stays on the manifold by construction, leaves all sixteen verdicts unchanged and narrows the environment null from 2.52 to 0.87 on CartPole, so the off-manifold draws were costing the test power rather than manufacturing it.

Acrobot also shows the test tracking *whether there is anything to explain* rather than agent quality: its agent never learns at this budget, scoring exactly -500 with a span of exactly 0.00 at every checkpoint and on five independent runs. Its environment reference is a point mass throughout, so the comparison is unresolved rather than negative, the same failure blinding shows on our synthetic corpora. CartPole’s observations matter at every skill level and the environment null fires throughout, from return 158 to 500, and it is the only one of the four tasks supplying a non-zero span at all. We do not claim undertrained agents generally produce empty explanations.

4.5 Attribution is not identified by task performance

We add a training penalty on the divergence between $\pi(\cdot | s)$ and $\pi(\cdot | s')$, where s' has one feature resampled from the batch marginal: the same perturbation the attribution measures (Table 4).

The attribution is equally suppressible in both cases. What differs is the price: suppressing the planted signal destroys 81% of return, while on the market no return difference is detected. A non-significant comparison is not evidence of equivalence, so we bound the effect in-

stead of asserting one. The market moves by $+1.84$ per episode, 95% interval $[-1.18, +5.46]$; the planted signal by -36.38 , interval $[-46.04, -18.98]$. *The intervals do not overlap*: the largest cost the market experiment cannot exclude is a sixteenth of the smallest cost consistent with the planted signal. Explanations are steerable without a detected cost exactly where they are not tracking anything, corroborating Section 4.1 by a route sharing no machinery with the null test. The control matters: had steering succeeded on both, the penalty would merely be defeating the attribution method.

This differs from Slack et al. [10], who construct classifiers that detect the off-manifold points LIME and SHAP query. Here the agent is trained normally and the resulting explanation is *correct* about the resulting model; the point is that task performance does not pin it down. On 4,096 held-out states the steered and unsteered agents agree on the greedy action only 66.0% of the time, so they are not the same policy and the claim is about performance rather than behaviour.

4.6 Three replacements, three failures

Our null trains its agents on synthetic signal-free corpora, so the corpus varies along with the information. Two constructions hold it fixed instead, and both fail.

Blinding the observation channel is what the definition above literally says. Run on every case it produces a false positive: on a corpus built to contain no informative feature, span $+13.90$ against a null of 0.00 ± 0.00 , which fires. Its verdict on the real market also flips with the null agents’ training budget, from $z = +0.55$ at 20 updates to $z = +203.6$ at 160, as those agents converge. Blinding gives agents that cannot see; they converge to a constant policy, whose return does not depend on its inputs at all, so every null span is exactly zero. The agent under test is sighted, so the right reference is sighted agents with nothing to learn.

Permuting the outcomes keeps them sighted and is the direct analogue of the supervised test. Observations are untouched to the bit and the outcome rate is preserved exactly. It has power ($z = +10.48$ on the planted signal) and specificity ($z = +1.75$ on the signal-free corpus) and still fails: on the real market the agent’s span lands *fifteen standard deviations below* its own null, because the null agents earn $+36.26$ an episode where it earns -1.13 . The price forecasts the outcome, so relabelling leaves the forecast attached to a different contract and the market becomes grossly mispriced. Fading the terminal price, a rule needing only the quote, earns -0.002 per contract in the real corpus and $+0.486$ in the permuted one. The null agents learn that arbitrage and it requires seeing the price. Wherever an observation forecasts the label, permuting the label makes the forecast exploitably wrong.

The pattern. Removing the information also perturbs something else, and the perturbation decides the verdict: the corpus, the agent’s responsiveness, or the

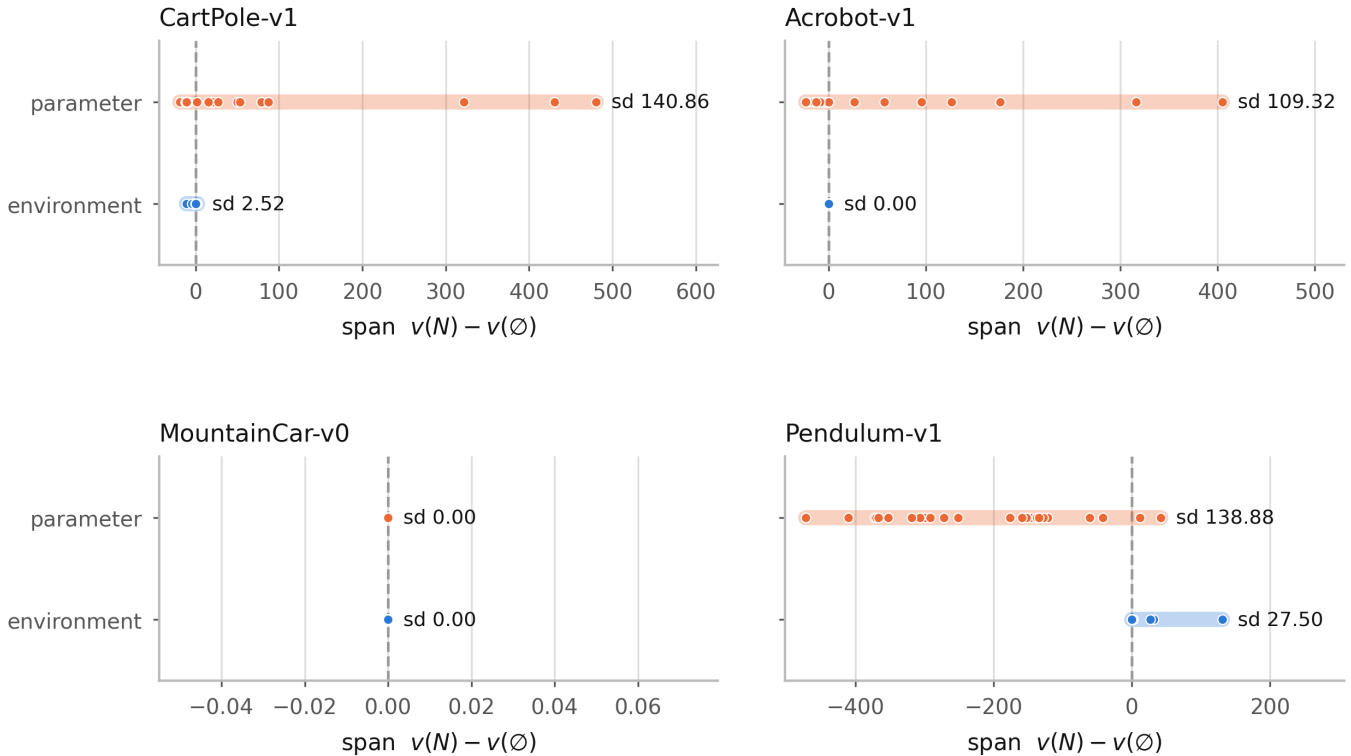


Figure 3: Spread of each null construction. Width determines power: a very wide reference distribution cannot reject anything.

difficulty of the task. We keep the signal-free-corpus null because its fault is the mildest and runs in the conservative direction, which is a reason to trust our negative results more than our positive ones.

5 Limitations

The test adapts Adebayo et al. [1]; the outcomes framing is Beechey et al. [3]’s. The span measures information in aggregate, not whether the decomposition among features is right. Temporal credit assignment and off-policy attribution are named, not solved; these are FastSVERL’s [2] territory. An earlier mechanism we proposed for the $56\times$ gap (that randomising weights perturbs the measurement’s domain and compounds along the trajectory) was tested and rejected: over horizons 4 to 28 both nulls grow at indistinguishable rates ($\alpha = 0.94$ against 0.83). A variance decomposition accounts for it instead, and involves no compounding. Four environments is too few to fit a law, so we report that correspondence rather than a coefficient. The aggregate verdict also survives a k -nearest-neighbour conditional masker that substantially reduces off-manifold states, though the per-feature ranking does not, disagreeing there on the most important feature. Accordingly the test validates whether there is something outcome-relevant to explain, not which feature attribution is correct. Scope is one real domain and four classic-control tasks, all with small observation spaces, so nothing here speaks to pixel-based agents.

6 Conclusion

An attribution can be stable, precisely estimated, plausible, and empty. The checks practitioners commonly apply do not detect this, because none of them asks whether there was anything outcome-relevant to explain. A null-model test asks it directly, using a statistic the attribution framework already provides. Building one is the hard part: every construction we tried removes the information and perturbs something else, and the perturbation decides the verdict.

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